

Northstar Operations Room

WYRESTORM | TECHNICAL PROPOSAL

Executive Summary

Route two operator sources to four displays with resilient 4K distribution.

This document sets out the WyreStorm recommendation for this requirement, how it will work in use, the equipment involved, and what remains to be confirmed before it is issued as a final customer proposal.

Customer Requirement

Confirmed requirement: Route two operator sources to four displays with resilient 4K distribution.

Discovery Conversation

The recommendation in this document is based on the discovery conversation captured below: the question asked, the closest governed answer, and the customer's own wording where recorded. Rows marked "Confirmed with customer" were verified with the customer during discovery; rows marked "To be confirmed" are still open and must be verified before final design sign-off. Nothing here is presented as a settled fact that was not said during discovery.

QUESTION ASKED	GOVERNED ANSWER	CUSTOMER WORDING	STATUS	CAPTURE CONFIDENCE
How many displays must receive independently routed content?	Four displays	Each operator needs independent routing.	Confirmed with customer	High confidence (12)

The "Capture confidence" column shows how strongly the customer's wording matched the closest governed answer: "High confidence" is a strong exact match, "Matched" is a curated-phrase match, and "Low confidence — verify before quote" is a partial match that must be re-verified with the customer before quoting. The number in brackets is the match score behind the tier.

Where a row is a note-only capture, the answer was still open at the time of writing and must be confirmed before final design sign-off.

Recommended Solution

Recommendation: **NetworkHD** is the leading product-family path for this requirement, based on the recommendation evidence gathered so far (family confidence 96/100).

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

Information required: Confirm redundant power and switch failover behaviour during commissioning.

Required WyreStorm products: NHD-600-TX - NetworkHD 600 encoder; NHD-600-RX - NetworkHD 600 decoder; NHD-CTL-PRO-V2 - NetworkHD controller

Practical Operation

This is what the system will do for the people using the room, day to day.

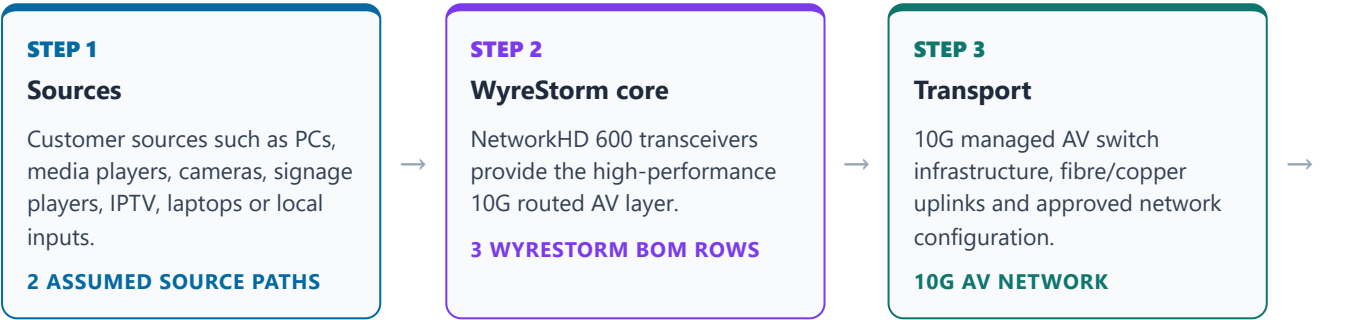
- Users will select the source they need and it will appear on the assigned display(s) without reconnecting any equipment.
- Routing runs over the AV-over-IP network, so additional displays, sources or rooms can be added later within the same architecture.
- Day-to-day operation is intended to use simple named presets or a control interface rather than requiring the user to understand the underlying signal path.

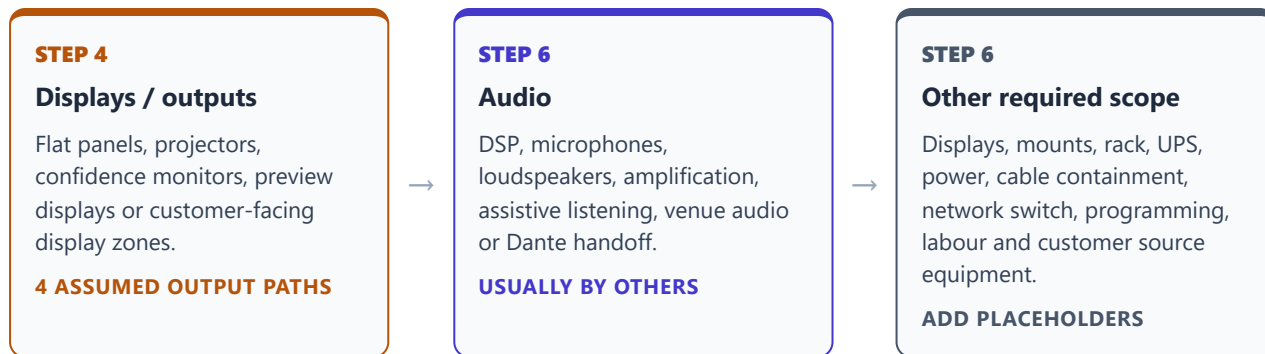
Commercial and Operational Benefits

- A consistent, repeatable connection experience across the room(s) covered by this proposal.
- An AV-over-IP architecture gives a clear, supported path to add displays, sources or rooms later without redesigning the system.
- Final commercial and operational benefit statements should be confirmed against the customer's specific business or KPI outcome before proposal issue.

Technical Architecture

The following diagram and cable schedule explain how the recommended solution connects together, derived directly from the equipment schedule below.





This diagram is derived directly from the equipment schedule below - it shows the assumed signal flow from source to display so the customer and installer share the same picture of how the room connects together. The cable schedule underneath lists the transport type, connectors, maximum length, and validation status for each connection.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
NHD-600-TX → NHD-600-TX	Video	15m	HDMI-A to HDMI-A	Confirmed	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-TX → NHD-600-TX	Video	15m	HDMI-A to HDMI-A	Confirmed	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-TX → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-TX → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
NHD-600-RX → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-RX → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-RX → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-RX → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-CTL-PRO-V2 → Network switch	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-RX → NHD-600-RX	Video	15m	HDMI-A to HDMI-A	Confirmed	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
NHD-600-RX → NHD-600-RX	Video	15m	HDMI-A to HDMI-A	Confirmed	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-RX → NHD-600-RX	Video	15m	HDMI-A to HDMI-A	Confirmed	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-RX → NHD-600-RX	Video	15m	HDMI-A to HDMI-A	Confirmed	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
Control system → NHD-600-TX	Control	100m	DB-9 / IR / RJ45	Confirmed	Confirm control protocol (RS-232/IR/IP), baud rate, pinout, cable routing, and control processor compatibility.
AV network infrastructure	Network / AVoIP	Unlimited	RJ45 to RJ45	Needs site confirmation	Confirm switch model, port count, PoE budget, VLAN configuration, multicast policy, and uplink capacity.
Control system integration	Control	100m	DB-9 / IR / RJ45	Needs site confirmation	Define who operates the room and convert engineering routes into user-friendly presets.

Equipment Schedule

ITEM	SKU	DESCRIPTION	ROLE	QTY	TYPE	NOTES
1	NHD-600-TX	NetworkHD 600 encoder	Encode operator sources	2	Required	Install in the equipment rack.
2	NHD-600-RX	NetworkHD 600 decoder	Decode display feeds	4	Required	One decoder per display.
3	NHD-CTL-PRO-V2	NetworkHD controller	Routing control	1	Required	Confirm firmware compatibility.

Evidence basis: NHD-600-TX: Two confirmed operator sources | NHD-600-RX: Four independently routed displays | NHD-CTL-PRO-V2: Central routing and control requirement

Power Strategy

The power budget below is derived from the governed technical profiles for each selected product. Figures are stated-maximum consumption per the published profile and must be verified against the current datasheet before order.

SKU	QTY	PER-UNIT MAX	TOTAL MAX	POWER DATA (GOVERNED PROFILE)
NHD-600-TX	2	Not proven	—	No power data in governed profile
NHD-600-RX	4	Not proven	—	No power data in governed profile
NHD-CTL-PRO-V2	1	15.4W	15.4W	12V DC PoE compliant; Max 15.4W, average 8W

Stated maximum consumption: approximately 15W across 1 product with proven figures. 1 product (NHD-CTL-PRO-V2) requires PoE/PoH — confirm the injector or switch PoE budget covers the total.

Options

- No optional enhancements are currently selected.

Assumptions and Exclusions

Assumptions

- Assumption:** Customer IT will provide a dedicated 10GbE AV VLAN.

Exclusions

- Display, projector, LED and mounting hardware supply unless listed as a specific line item above.

- Electrical, structural and builder's work.
- Network switch procurement, VLAN/multicast configuration and IT commissioning access.
- Structured cabling installation and certification.
- Third-party control programming beyond WyreStorm-native presets.
- Installation, labour and commissioning, unless separately quoted.

Risks and Dependencies

Risks / information required

- **Risk:** Confirm redundant power and switch failover behaviour during commissioning.

Governed dependencies

- **Dependency: NET-SWITCH-10G-MANAGED - Managed 10GbE network switch** (Prompt): Confirm switch model, port budget, VLAN and multicast configuration. Basis: NetworkHD endpoints require managed 10GbE switching.

Implementation and Next Steps

- Approve the network design and complete acceptance testing.

This proposal draft includes selected WyreStorm products. Validate final product specifications, accessories and dependencies before issue.

Competitor products are excluded from this proposal and equipment schedule unless a comparison-only output is explicitly requested.

Important Notice & Best-Efforts Disclaimer

This proposal has been prepared on a best-efforts basis using WyreStorm Wingman to support your sales and design process. The information contained herein is provided for planning purposes only and does not constitute a binding quotation, order confirmation, or statement of capability.

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Technical Validation: System designs, power budgets, cable schedules, and equipment selections shown in this proposal are based on available product data and assumptions stated within. Final designs must be validated by a qualified AV integrator or

pre-sales engineer before installation.

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Northstar parity evidence fixture

Confirmed Requirement

Route two operator sources to four displays with resilient 4K distribution.

Design Assumptions

- Customer IT will provide a dedicated 10GbE AV VLAN.

Recommended Architecture

NetworkHD is the current leading product-family direction.

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

Required WyreStorm Products

- **NHD-600-TX** - NetworkHD 600 encoder
- **NHD-600-RX** - NetworkHD 600 decoder
- **NHD-CTL-PRO-V2** - NetworkHD controller

Optional Enhancements

- No optional enhancements are currently selected.

Risks / Needs Validation

- Confirm redundant power and switch failover behaviour during commissioning.

Next Steps

- Approve the network design and complete acceptance testing.